FLOODSENSE: AN ANDROID-BASED EXPERT SYSTEM FOR FLOOD SUSCEPTIBILITY ASSESSMENT AND PRE-EVENT PREPAREDNESS IN BACOOR CITY

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FloodSense: An Android-Based Expert System for Flood Susceptibility Assessment and Pre-Event Preparedness in Bacoor City

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INTRODUCTION

Flooding remains one of the most persistent hazards affecting Philippine communities. Its consequences extend beyond inundation and property damage to interrupted travel, disrupted livelihoods, displacement, and difficult household decisions before and during severe weather. National disaster statistics continue to document the effects of extreme events on Filipino families and communities (Philippine Statistics Authority [PSA], 2026). These conditions make accurate, understandable, and appropriately qualified risk information an essential part of disaster preparedness.

Bacoor City, Cavite, includes low-lying and river-connected areas exposed to recurring flood conditions. Flood-mitigation works in the Imus River Basin were established to reduce peak floodwater volume in Cavite, including low portions of Bacoor and its tributaries (Japan International Cooperation Agency [JICA], 2021). Local interviews also identified historical flood records, geographic materials, preparedness procedures, and evacuation information held by city offices, together with restrictions on the availability and public use of machine-readable spatial data.

Residents may obtain flood information through official social-media pages, barangay announcements, text messages, broadcast media, websites, community networks, and personal observation. However, the ongoing resident survey must still establish how eligible respondents assess the accessibility, timing, clarity, location specificity, consistency, and usefulness of these sources. [Insert final number of valid respondents.] [Insert final percentage of respondents who reported this problem.] Preliminary survey reports retained in the project are treated only as earlier needs-assessment records and are not pooled with the active survey or presented as final findings.

Research on flood-risk communication emphasizes that technical information is most useful when users can understand its geographic meaning, source, limitation, and connection to appropriate preparedness action (Mostafiz et al., 2022). Geographic provenance is also necessary because a technically processed boundary or map does not automatically become institutionally verified hazard information (Tullis & Kar, 2021). In Bacoor City, the absence of an approved operational susceptibility dataset and the need for local expert validation remain central constraints.

Against this context, the study examines a scenario-based approach to pre-event flood-susceptibility assessment and preparedness support. The approach keeps a deterministic assessment separate from decision-support guidance, requires the user to confirm a hypothetical rainfall scenario and location, and discloses that results are not live observations, official forecasts, warnings, or evacuation orders.

FloodSense is developed as an Android resident application and a separate Web Administration System that organize these verified concepts. The current prototype demonstrates rule-based assessment, explanation, interactive geographic presentation, structured preparedness guidance, resident account controls, and governed administrative workflows. Its public operational validity remains dependent on approved local data, documented provenance, expert and institutional validation, completed evaluation, and production arrangements.

Statement of the Problem

Flood preparedness in Bacoor City depends not only on the availability of information but also on whether residents can obtain it in time, understand it, relate it to a location, and use it in an appropriate decision. City offices hold relevant historical, geographic, preparedness, and evacuation information, yet access restrictions, data provenance, validation, and public communication responsibilities limit what can be translated directly into a public assessment. The active resident survey is still collecting evidence about current sources, timing, clarity, location specificity, consistency, information needs, and technology access. [Insert final number of valid respondents.] [Insert final distribution by barangay.]

The local research problem therefore has two connected dimensions. The first is an evidence problem: the study must establish, without treating incomplete responses as final, which information difficulties and preparedness needs eligible respondents actually report. The second is a governance and design problem: any scenario-based susceptibility assessment must use traceable and locally appropriate knowledge, distinguish administrative reference geography from flood-susceptibility data, explain its basis and limitations, and remain separate from official forecasting and warning authority.

The main problem of the study is how verified resident needs, local institutional requirements, controlled geographic and rainfall information, deterministic rule-based inference, and source-backed preparedness guidance can be organized into an understandable pre-event support system for Bacoor City without overstating the authority or completeness of its data.

Specifically, the study seeks to answer the following provisional questions: (1) What flood-information practices, access conditions, and problems do eligible Bacoor City respondents report? (2) What verified knowledge, rainfall parameters, geographic records, preparedness content, and governance controls are required for a locally

appropriate scenario-based assessment? (3) How can the verified requirements be implemented through an Android resident application and a separate Web Administration System? (4) To what extent do the implemented functions operate correctly and consistently under defined technical test cases? (5) How do the defined evaluator groups assess the relevant FloodSense product-quality characteristics using an approved ISO/IEC 25010-based instrument? Final wording will be reviewed after the ongoing survey is closed and analyzed.

Objectives of the Study

Generally, the study aimed to develop and evaluate FloodSense as a scenario-based Android system for explainable flood-susceptibility assessment and separate pre-event preparedness support for Bacoor City, subject to validated local knowledge, data provenance, and institutional limitations.

Specifically, it aimed to:

1. identify the flood-information practices, access conditions, problems, and preparedness needs reported by eligible Bacoor City respondents and relevant local offices through the active resident survey, interviews, and document review;
2. determine the locally appropriate facts, rainfall parameters, geographic records, rule conditions, preparedness content, evacuation-center records, sources, validation states, and governance controls required for the knowledge base and supporting services;
3. develop the Flutter-based Android resident application and Django-based Web Administration System with: (a) resident authentication, legal acceptance, onboarding, preferences, and account controls; (b) user-confirmed rainfall and location scenarios; (c) deterministic rule matching, conflict handling, susceptibility output, and explanation; (d) provider-aware interactive mapping, manual selection, map pin, one-

time foreground location, and barangay confirmation; (e) structured Decision Support System guidance kept separate from the assessment; (f) nearest-center information subject to authorized current records; and (g) governed administrative workflows that do not expose raw rule editing to ordinary administrators;

4. test the implemented mobile, backend, geographic, Expert System, Decision Support System, account, map, permission, and administrative functions through defined unit, integration, API, spatial, and end-to-end test cases; and
5. evaluate the relevant product-quality characteristics of FloodSense using an approved ISO/IEC 25010-based instrument and defined evaluator groups. [Insert final evaluation participants, scores, and interpretation after data collection.]

Theoretical Framework

The theoretical and conceptual framework of FloodSense relates user-confirmed inputs, controlled knowledge, deterministic inference, geographic presentation, and separate preparedness support. The framework uses four function-named modules so that technologies and database storage remain implementation mechanisms rather than conceptual modules.

Scenario and Location Input Module. The resident selects or confirms a hypothetical rainfall intensity and duration, chooses a supported location manually or through an optional one-time foreground location request, and confirms or corrects the proposed barangay before explicitly requesting an assessment. These inputs are not live measurements or forecasts.

Susceptibility Assessment and Explanation Module. The deterministic Rule-Based Expert System validates the confirmed inputs, matches applicable rules, resolves priorities or conflicts, and returns a Low, Moderate, High, or Very High class

with a human-readable explanation. Repeatability does not establish scientific validity; the rules and local data still require provenance, expert review, and institutional authorization.

Preparedness and Evacuation Information Module. A separate structured Decision Support System uses the assessment only as context for published guidance and does not change its class or issue an evacuation order. The mobile application can request nearest-center information, but distance does not establish route safety, opening status, accessibility, or live capacity, and public use depends on authorized current records.

Governance and Data Stewardship Module. Authorized administrators use the custom Web Administration System for dashboards and controlled workflows involving map and rainfall references, parameters, DSS content and structured flows, evacuation centers and staged center imports, data sources, and audit history. Ordinary administrators do not edit raw rules, rule conditions, priorities, conflict-resolution logic, or algorithm code. Technical Django Admin remains separate. The confirmed fixed-schema Expert System parameter CSV exchange is not yet implemented.

The four modules produce an explainable, scenario-specific result, geographic presentation, and preparedness information while maintaining source, permission, validation, and limitation boundaries. The 47-barangay layer is an administrative reference rather than a susceptibility dataset, and the framework does not include real-time monitoring, autonomous alerts, official forecasts or warnings, background GPS, automatic evacuation orders, safe-route calculation, machine-learning prediction, or self-modifying rules.

[AUTHOR NOTE: Insert the approved Conceptual Framework diagram here after the provisional module names and relationships are reviewed.]

The provisional System Architecture places the resident on the left, the authorized administrator on the right, and the resident application, custom Web Administration System, Django REST services, deterministic inference, geographic services, structured DSS, and evacuation-center service between them. PostgreSQL/PostGIS provides controlled relational and spatial storage below these services. Approved rainfall, geographic, preparedness, and center information enters as external reference data rather than as users or actors. Provider-aware Mapbox/OSM interfaces present maps when configured. [AUTHOR NOTE: Insert the approved System Architecture diagram here after actor and data-source validation.]

Significance of the Study

FloodSense investigates how an explainable, scenario-based assessment and separate preparedness guidance may support pre-event understanding in Bacoor City. Any benefit described below remains subject to approved data, completed evaluation, and the system's role as a complementary information tool.

For the Bacoor Disaster Risk Reduction and Management Office (BDRRMO). The governed prototype may provide an additional channel for publishing validated preparedness and evacuation-center information while preserving source and audit records. It does not replace official warnings, operations, or emergency directives.

For the City Planning and Development Coordinator Office (CPDCO). The study demonstrates how appropriately authorized planning and geographic information may be referenced in a location-specific application. City-held data cannot be assumed public or operational without permission, provenance, and validation.

For Bacoor City residents. The resident application provides a user-initiated flow for confirming a hypothetical rainfall scenario and location, requesting an explained susceptibility result, viewing a map, completing structured preparedness guidance, and viewing nearest-center information when verified records are available. It does not provide live monitoring, official forecasts or warnings, safe-route advice, live occupancy, or safety guarantees. [Confirm whether the survey evidence supports retaining this feature.]

For future researchers. The study documents the separation of deterministic assessment from preparedness guidance, the role of geographic provenance and validation, and the difference between implemented software and authorized operational data. Its methods and findings should not be transferred to another locality without renewed validation.

Time and Place of the Study

The study is scheduled from September 2026 to May 2027. Development, testing, consultation review, and manuscript preparation are conducted primarily at Cavite State University–Bacoor City Campus, Soldiers Hills IV, Molino, Bacoor City, Cavite.

Resident survey activities are conducted in included Bacoor City communities. Recorded interviews and document requests involve the BDRRMO at the City of Bacoor Legislative and Disaster Resilience Building and the CPDCO at the Bacoor Government Center, Barangay Bayanan. [Insert final survey distribution locations and dates.]

Scope and Limitation of the Study

FloodSense is a scenario-based, pre-event flood-susceptibility assessment and preparedness-support system intended for Bacoor City residents and authorized administrators. A resident authenticates, completes the required legal and onboarding

steps, selects or confirms a hypothetical rainfall intensity and duration, selects or confirms a supported location, and explicitly requests an assessment. The deterministic Rule-Based Expert System returns an explained Low, Moderate, High, or Very High demonstration class, while a separate Decision Support System presents source-backed preparedness guidance without changing that class. The resident application supports manual area selection, map-pin adjustment, optional one-time foreground location, barangay confirmation or correction, interactive geographic display, account and preference controls, and nearest-center lookup when authorized and current center records are available; however, the location request does not continuously track the device, and a displayed center or distance does not verify road passability, accessibility, current opening, available capacity, or safety. The custom Web Administration System provides authenticated dashboards and governed workflows for map and rainfall references, parameter review, DSS content and structured flows, evacuation-center records and staged center import, data sources, and audit history. Ordinary administrators cannot modify raw Expert System rules, rule conditions, priorities, conflict-resolution logic, or algorithm code, and the separate technical Django Admin is reserved for developer or research maintenance; however, the confirmed fixed-schema CSV export and import workflow for permitted Expert System parameter values remains unimplemented and is distinct from the available evacuation-center import workflow. The 47-barangay layer provides administrative reference and has undergone technical processing for software use; however, it is not automatically City-issued or City-verified, contains no flood-susceptibility facts by itself, and no approved operational Bacoor City susceptibility dataset is established. FloodSense does not provide real-time rainfall or water-level monitoring, automatic rainfall updates, background polling, background GPS tracking, autonomous alerts, official forecasts or warnings, automatic evacuation orders, safe-road routing, live center occupancy, machine-learning prediction, self-modifying rules, complete offline operation, resident assessment history, completed evaluation, or verified production

deployment. Provisional geographic materials, processing scripts, mockups, screenshots, and technically valid geometries do not establish institutional approval or operational integration. Automated tests provide evidence of software behavior within their test cases, but they do not establish scientific validity, institutional approval, usability, or production readiness. Personal information and one-time foreground location must be handled under Republic Act No. 10173 and the approved consent and privacy process, and direct identifiers from the ongoing survey must not be included in the analysis dataset or public results.

Definition of Terms

The following terms are defined operationally as they are used in this study to ensure clarity and consistency throughout the research:

Android Application. The resident-facing FloodSense software developed with Flutter. It includes authentication, legal acceptance, onboarding, scenario and location confirmation, assessment requests, map/results presentation, structured preparedness guidance, nearest-center information, preferences, and account controls. It does not provide live alerts, assessment history, fully offline operation, or official warnings.

Check My Area. The location-selection flow that allows manual area selection, placement or adjustment of a map pin, and an optional one-time foreground location request. A detected barangay is proposed for user confirmation or correction before assessment; the function does not continuously track the device.

Color-Coded Susceptibility Levels. The visual scheme used on the dynamic map in which green indicates Low, yellow indicates Moderate, orange indicates High, and red indicates Very High flood susceptibility for the active scenario.

CPDCO (City Planning and Development Coordinator Office). The Bacoor City office that maintains planning and geographic information, including elevation, land-use, flood-hazard, and barangay-boundary materials relevant to the study.

Decision Support System (DSS). The component separate from the Rule-Based Expert System that presents published preparedness guidance and structured questions using the assessment context. It does not change the susceptibility result, issue an official warning or evacuation order, or make the user's final decision.

Dynamic Map. The provider-aware interactive map that supports panning, zooming, scenario-dependent demonstration coloring, manual selection, map-pin placement, foreground location, reference boundaries, and center markers when data are available. Its display does not convert provisional data into official hazard information.

Expert-Provided and Expert-Validated Knowledge. Flood-related facts, parameters, rules, preparedness guidance, and supporting information that have been supplied or reviewed by an authorized source and recorded with adequate provenance. Repository presence or technical validation alone does not establish institutional approval.

Expert System. A knowledge-based computer system that applies stored facts and conditional rules through an inference process. In FloodSense, it produces an explainable susceptibility class for a supported geographic area and rainfall scenario.

Flood Susceptibility. The relative potential of a supported geographic area to experience rainfall-related flooding under a user-confirmed scenario. The current prototype demonstrates Low, Moderate, High, or Very High outputs but does not yet contain an approved operational Bacoor City susceptibility dataset and is not a real-time prediction.

Geographic Information System (GIS). A system for organizing, analyzing, and presenting information associated with geographic locations. FloodSense uses GIS functions for polygons, points, containment queries, interactive visualization, and evacuation-center locations.

GeoDjango. The geographic extension of Django used by FloodSense to represent spatial data and perform geographic operations between backend services and the PostGIS database.

Inference Process. The procedure that matches facts from the selected rainfall scenario and geographic area against validated rules in the knowledge base to derive a susceptibility conclusion and supporting explanation.

Knowledge Base. The structured collection of expert-validated facts, conditional rules, geographic attributes, rainfall parameters, susceptibility categories, and related provenance used by the Expert System.

PAGASA-Aligned Rainfall Input. The current interface labels for Light, Moderate, Heavy, Intense, and Torrential rainfall and durations of 1, 3, 6, 12, or 24 hours. These selections are hypothetical scenario inputs. Their operational thresholds and use require authoritative source confirmation and validation before official deployment.

PostGIS. The spatial extension of PostgreSQL used to store and query FloodSense geographic polygons, points, boundaries, selected locations, and evacuation-center coordinates.

Preparedness Provenance. Information identifying the source, validation status, responsible actor, relevant date, and version of a preparedness item or related record. Provenance permits users and administrators to distinguish approved, provisional, and unsupported information.

Rainfall Parameters. The supported combination of rainfall intensity and duration selected by the user and applied as scenario facts during Expert System assessment.

Rule-Based Expert System. An Expert System in which domain knowledge is represented as conditional rules. FloodSense evaluates applicable rules against known scenario and geographic facts to produce a susceptibility assessment.

Web Administration System. The custom browser-based interface used by authorized administrators for dashboards and governed workflows involving map and rainfall references, parameters, DSS content and structured flows, evacuation centers and staged center imports, data sources, and audit history. It does not expose raw rule editing. Technical Django Admin is separate. Fixed-schema Expert System parameter CSV export/import is confirmed but not implemented.

REVIEW OF RELATED LITERATURE

This chapter presents a review of related literature, articles, and studies that provide guidance in identifying and formulating the requirements needed to start the project. Moreover, this chapter was designed to examine relevant concepts, methods, challenges, and practices associated with flood-susceptibility assessment, geographic information systems, Rule-Based Expert Systems, and Decision Support Systems.

Related Foreign Literature

Flood Susceptibility and Scenario-Based Assessment

Flood susceptibility expresses the relative tendency of a location to experience flooding when relevant conditions are present. It differs from flood hazard, which concerns the probability and physical characteristics of an event; exposure, which concerns people or assets in affected places; vulnerability, which concerns the capacity to be harmed; and risk, which combines hazard and potential consequences. Membele, Naidu, and Mutanga (2022) found that flood-vulnerability mapping in developing countries varies substantially in terminology, indicators, spatial scale, data availability, and validation practice. Their review shows why a public system must name the type of assessment it performs and avoid presenting a general vulnerability or hazard map as a locally validated susceptibility conclusion.

Scenario-based assessment supports planning by holding selected conditions explicit instead of implying that the system is observing an event as it occurs. A hypothetical rainfall intensity and duration can be treated as assessment inputs only when their meanings, permitted combinations, and geographic applicability are documented. This supports the FloodSense interaction in which the resident selects or confirms a scenario before requesting assessment. It does not establish that the scenario is current, predicted, or officially issued, and it does not justify importing weights or thresholds derived for another place.

Deterministic Rule-Based Expert Systems and Knowledge Bases

A Rule-Based Expert System represents domain knowledge through facts, conditions, and production rules and applies an inference procedure to derive a conclusion. Buchanan and Shortliffe (1984) provide the foundational distinction between the knowledge base and the inference mechanism. More recently, Papadopoulos et al. (2022) reviewed technologies and standards used in rule-based decision-support systems and described how encoded knowledge and rules engines evaluate case-specific facts. Although their review concerns clinical systems, the architectural distinction is relevant to FloodSense: flood knowledge, scenario facts, geographic facts, and explanations belong to controlled knowledge structures, while the inference procedure determines which eligible rule applies.

Deterministic processing means that the same approved facts and the same published rule version should produce the same result. It does not imply that the rules are scientifically valid merely because the software executes them consistently. Knowledge acquisition, conflict handling, parameter definitions, version control, and expert or institutional validation remain necessary. Accordingly, ordinary FloodSense administrators may manage approved records and permitted values but do not rewrite raw rule conditions, priorities, conflict-resolution logic, or algorithm code. The confirmed fixed-schema CSV exchange is a future controlled-value workflow and is not evidence that such exchange is currently operational.

Explainable Inference

Explainability is necessary when a system produces a classification that may influence preparedness decisions. Kostopoulos, Davrazos, and Kotsiantis (2024) reviewed explainable decision-support approaches and emphasized transparency, interpretability, and human-understandable justification. Their review primarily addresses artificial-intelligence-based systems, but its communication principles apply to deterministic inference: a user should be able to understand the relevant inputs, the

basis of the result, and the limitations of the evidence without being required to inspect source code or raw rule definitions.

For FloodSense, an explanation should identify the confirmed rainfall scenario, resolved geographic context, susceptibility class, applicable basis, source or version information, and any limitation state. This supports accountability while preserving the distinction between explanation and rule administration. It also prevents a color or label from appearing authoritative without disclosing what it represents. Explainability does not remove the need for validated data, and a clear explanation cannot turn provisional information into an official local assessment.

Decision Support and Preparedness Guidance

A Decision Support System organizes information, alternatives, or recommended actions to assist human judgment. In flood preparedness, support may include understandable mitigation choices, household preparation, communication planning, protection of documents and valuables, assistance for vulnerable household members, attention to official advisories, and evacuation readiness. Mostafiz et al. (2022) observed that many web-based flood tools provide technical information but do not consistently provide locally actionable information for individuals and communities. Their review supports combining understandable risk communication with practical guidance while clearly stating the tool's authority and limitations.

FloodSense separates the susceptibility assessment from preparedness guidance. The deterministic Expert System produces the assessment, while the DSS presents source-backed pre-event guidance without changing the susceptibility class. Information about evacuation facilities may support readiness when its source, verification date, location, and operational limitations are known, but distance alone does not prove route safety, current opening, accessibility, or available capacity. The DSS therefore assists preparation and does not issue an official warning, evacuation order, or guarantee of safety.

GIS, Interactive Maps, and Spatial Databases

Geographic Information Systems organize, analyze, and present information associated with locations. Their value in flood research depends on the suitability of the layers, coordinate reference systems, spatial resolution, classification methods, and validation procedures. An administrative boundary can resolve the name of a barangay or provide map context, but it contains no flood-susceptibility fact unless appropriate hazard or susceptibility evidence has been associated with it. This distinction is important when users may interpret any colored polygon as an official hazard layer.

Spatial databases support the controlled storage and querying of geographic records. The PostGIS Development Group (2026) explains that PostGIS extends PostgreSQL with geometry and geography types, spatial indexes, spatial relationships, coordinate transformation, validity operations, and spatial processing. These capabilities are relevant to point-in-polygon queries, administrative-area lookup, map-feature storage, and provenance-linked geographic records. They provide technical infrastructure, not scientific validation: a valid geometry and a successful spatial query do not establish that a flood conclusion is institutionally approved.

Flood-Risk Communication

Flood-risk information must be understandable, actionable, and explicit about uncertainty. Mostafiz et al. (2022) identified gaps in web-based flood communication, particularly where tools provide information without sufficiently connecting it to decisions that residents can make. Clear terms, legends, labels, explanation text, and source notices help prevent non-specialists from treating model outputs as certain predictions. Communication should also distinguish long-term or scenario-based information from time-sensitive official warnings.

These principles support a guided FloodSense flow rather than an unexplained technical map. Susceptibility colors must be paired with text, the selected scenario,

and limitations. Preparedness information must direct users toward official channels during actual events. The interface must not imply that a neutral boundary layer, a demonstration color, or a provisional record represents a current flood condition.

Data Provenance and Geographic Validation

Provenance records where information originated, how it was transformed, which version was used, and what review or approval applies. Tullis and Kar (2021) connected provenance to ethical replicability and reproducibility in GIScience, including critical applications in which geographic data affect public decisions. Fakhruddin et al. (2022) similarly emphasized risk-informed data practices and the importance of understanding how disaster and climate data are produced, shared, interpreted, and governed.

For FloodSense, provenance should accompany susceptibility parameters, geographic layers, preparedness items, and resource information. Technical geometry validation may confirm coordinate-system consistency, valid shapes, or feature counts, but institutional validation concerns authority, currency, meaning, and permitted public use. The two forms of validation are related but not interchangeable. This distinction prevents a technically processed dataset from being represented as official Bacoor City flood information without documented approval.

Foreground Location and Privacy

Mobile location can reveal a person's position and may contribute to identification or tracking. Android Developers (2026) distinguishes foreground access, one-time access, and background access and recommends requesting only the permission needed for the active function. This supports a FloodSense flow in which location is obtained only after a user initiates location selection or assessment, while manual selection remains available when the resident declines permission or prefers another area.

Foreground acquisition does not imply continuous monitoring. FloodSense does not require background GPS tracking, automatic polling, or autonomous location-based alerts. Location handling should remain purpose-limited to the active assessment, use only the precision necessary for the supported geographic operation, and avoid retaining unnecessary movement information. The presentation must also allow the user to confirm or correct the detected barangay because device accuracy and boundary proximity can affect geographic resolution.

Software Testing and Quality Evaluation

Software testing establishes whether specified behavior is implemented under defined conditions, while quality evaluation examines broader product characteristics. ISO/IEC 25010:2023 defines a product-quality model with nine characteristics for specifying and evaluating information and communication technology products (International Organization for Standardization [ISO], 2023). The model can guide the selection of criteria for FloodSense, but it does not provide a ready-made acceptance result or replace an approved evaluation instrument.

A defensible evaluation must identify the tested build, participants or evaluators, tasks, criteria, scale, procedure, valid responses, and interpretation method. Repository tests and interface demonstrations can support implementation verification, but they are not resident-acceptance findings. Because the final FloodSense survey and evaluation are not complete, Chapter 2 uses ISO/IEC 25010 only as literature supporting future assessment and does not report quality scores.

Existing Foreign Flood-Information Platforms

Foreign platforms demonstrate several forms of flood information without constituting transferable evidence for Bacoor City. FloodAdapt uses SFINCS-based scenario simulation and Delft-FIAT impact analysis to compare flood events and adaptation options (Deltares, 2025). The Global Flood Awareness System applies

large-scale hydrological modeling to operational river-flood forecasting (Copernicus Emergency Management Service, 2026), while the Iowa Flood Information System combines Iowa-specific observations, radar rainfall, forecasting, and mapped information (Iowa Flood Center, 2024). These services illustrate scenario analysis, forecasting, and public communication, but their models, data, scales, and authority differ from FloodSense.

The Resilience Analysis and Planning Tool combines United States hazard, infrastructure, demographic, and community-resilience layers for emergency-management planning (Federal Emergency Management Agency, 2025). England's National Assessment of Flood and Coastal Erosion Risk similarly communicates modeled long-term risk from several flood sources at national scale (Environment Agency, 2024). These official resources show the value of visible limitations, data context, and geographic lookup; however, they are platforms and institutional assessments rather than FloodSense research studies, and their datasets and classifications cannot validate a Bacoor City susceptibility result.

Related Local Literature

Philippine Flood Conditions and Institutional Preparedness

Official Philippine statistics continue to record substantial effects from extreme events and disasters. The Philippine Statistics Authority (PSA, 2026) summarizes national information on disaster occurrence, affected populations, deaths, and economic losses. These national figures establish the continuing relevance of disaster preparedness but do not identify the susceptibility of a particular Bacoor barangay or validate the inputs of a local application.

The National Disaster Risk Reduction and Management Council's National Disaster Response Plan identifies institutional responsibilities and coordinated actions for hydro-meteorological hazards (NDRRMC, 2024). The plan reinforces that warnings, response decisions, and emergency coordination belong to authorized government mechanisms. FloodSense may support pre-event understanding and

preparedness, but residents must continue to follow PAGASA, local disaster offices, barangay authorities, and emergency responders during an actual event.

Official Hazard Information and Warning Authority

PAGASA operates rainfall and flood-information processes that use observations, forecasts, water-level information, and coordinated warning protocols. Its 2023 Annual Climate Bulletin explains the role of rainfall warning systems and the use of official warning levels for public action (PAGASA, 2024). FloodSense rainfall selections are hypothetical assessment inputs; they are not automatic updates from PAGASA, an interpretation of current rainfall, or an official warning product.

GeoRisk Philippines presents HazardHunterPH as a national platform for site-specific indicative multi-hazard assessment using official information from participating government agencies (GeoRisk Philippines, 2023). Its source notices and limitations demonstrate responsible hazard communication. The same principle applies to FloodSense: sources, dates, limitations, and validation status should remain visible, and an indicative or provisional record should not be represented as an absolute prediction.

Cavite, Bacoor City, and the Imus River Basin

The Japan International Cooperation Agency reported the inauguration of flood-mitigation facilities associated with the Imus River Basin and low-lying areas of Imus and Bacoor (JICA, 2021). This institutional record confirms the relevance of watershed and infrastructure conditions to the local context. It does not provide barangay susceptibility classifications, rainfall-rule thresholds, or validation for FloodSense, and it should not be treated as a substitute for hydrologic or hazard data.

Administrative geography in Bacoor has also changed. The PSA documented the 2023 merger and renaming of barangays and currently lists 47 barangays in the City of Bacoor (PSA, 2023, 2025). These records support the current identities used

in a reference layer. They do not establish the accuracy of every derived polygon, City verification of a third-party boundary, or flood conditions within any barangay. Administrative reference and flood-susceptibility evidence must therefore remain separate.

Administrative Data Governance and Responsible Use

Government publications, local records, and research datasets differ in authority and permitted use. An official list of administrative units may verify names and codes, while an authorized hazard agency may provide hazard information, and a local office may validate operational records. Combining these sources requires documentation of ownership, date, scale, processing, approval, and restrictions. Provenance is particularly important when a dataset has been transformed, because technical processing can change geometry or attributes without changing the institution responsible for the original information.

FloodSense consequently distinguishes approved, provisional, technically validated, and unsupported information. The 47-barangay layer may serve as a current administrative reference after documented derivation and geometry checks, but it is not a flood-susceptibility layer and is not automatically City-issued or City-verified. No operational susceptibility dataset should be described as established until the necessary source, method, expert review, institutional authority, and database integration are documented.

Philippine Data Privacy and Software-Quality Context

Republic Act No. 10173, the Data Privacy Act of 2012, requires personal-information processing to observe transparency, legitimate purpose, and proportionality. For a resident-facing location function, these principles support clear notice, user initiation, limited collection, secure handling, and avoidance of continuous background tracking. They also support role-controlled administrative access. The Act

is legally authoritative despite falling outside the preferred publication period for recent literature.

Philippine software studies have also used ISO/IEC 25010 as an evaluation framework, but the criteria and edition must be identified accurately. An evaluation of another application cannot be transferred to FloodSense, and the presence of a questionnaire does not establish completed results. FloodSense quality conclusions will require its own approved instrument, respondents or evaluators, documented procedure, and completed analysis.

Existing Philippine Hazard-Information and Planning Platforms

UP NOAH remains an important Philippine reference for accessible geographic hazard information. Lagmay et al. (2017) documented the dissemination of near-real-time hazard information and flood maps through Web-GIS using open-source code and application programming interfaces. The dated publication supports discussion of public map-based risk communication, while the general platform should not be treated as a research study with an invented SDLC. UP NOAH differs from FloodSense because its documented hazard-information resources were institutionally developed, whereas FloodSense is a scenario-based research prototype and is not an official forecasting or warning platform.

HazardHunterPH is a GeoRisk Philippines application that produces site-specific indicative multi-hazard assessments from official agency information (GeoRisk Philippines, 2023). Its rapid geographic lookup and source-based reporting are relevant to FloodSense's emphasis on accessible information, provenance, and visible limitations. It remains an official platform rather than a substitute for locally validated FloodSense rules or data, and its reports do not confer forecasting or warning authority on FloodSense.

GeoMapperPH supports governed collection of hazard, exposure, vulnerability, and coping-capacity information by trained government and agency users, while

PlanSmartPH Ready to Rebuild organizes information for local-government rehabilitation and recovery planning (DOST-PHIVOLCS, 2022a, 2022b). These platforms illustrate structured data governance and institutional workflows. They are discussed as existing Philippine systems, not retained as Related Studies, because the inspected official pages do not provide a complete research methodology and results section suitable for the comparative study tables.

Related Foreign Studies

Examining Flood Vulnerability Mapping Approaches in Developing Countries: A Scoping Review

Membele, Naidu, and Mutanga (2022) examined how flood-vulnerability mapping was conducted in developing countries. Following PRISMA-ScR procedures, they searched Web of Science and Scopus for 2010-2020 publications, screened 2,549 records, and reviewed 130 eligible studies. They found frequent use of GIS with indicator-based and multicriteria methods, but limited stakeholder participation, sensitivity analysis, and validation, together with recurring problems involving coarse or outdated data.

The review is relevant because it shows that mapping terminology, indicators, spatial scale, participation, and validation must be reported before a flood-related output can be interpreted responsibly. Like FloodSense, the reviewed work concerns geographically organized flood information; however, it synthesizes vulnerability-mapping studies rather than building a resident scenario-assessment application. Its findings identify methodological weaknesses but do not supply Bacoor-specific factors, weights, rules, or validation, and the review was limited to studies that produced a flood-vulnerability map.

Actionable Information in Flood Risk Communications and the Potential for New Web-Based Tools for Long-Term Planning for Individuals and Community

Mostafiz, Rohli, Friedland, and Lee (2022) investigated how flood-risk information could better support long-term decisions by individuals and communities.

They reviewed flood communication through mass media, citizen science, and social media and examined 50 freely available web-based flood-risk portals. Their review found that many tools emphasize short-term information, while micro-scale, actionable, long-term planning information and explicit cost-benefit support remain limited.

The study supports FloodSense's use of understandable result explanations and separate preparedness guidance. Both emphasize that technical flood information must be connected to actions a user can understand; however, the study evaluated a cross-section of portals rather than a deterministic Philippine susceptibility system. Its portal sample was a time-bounded snapshot, and its findings cannot establish what information Bacoor residents need without local survey and stakeholder evidence.

Accelerated and Interpretable Flood Susceptibility Mapping Through Explainable Deep Learning with Hydrological Prior Knowledge

Wang, Sanderson, Iqbal, and Woo (2025) developed and evaluated an explainable deep-learning approach for flood-susceptibility mapping in a 1,516-square-kilometer area of Northumberland, United Kingdom. They combined official flood records and terrain and environmental variables, compared support-vector-machine, decision-tree, random-forest, standard U-Net, and hydrology-aware U-Net models, used spatial five-fold cross-validation, and applied Grad-CAM interpretation. The proposed model achieved a reported AUC of 0.966 and faster convergence than the standard U-Net.

The study demonstrates how susceptibility models can be evaluated and interpreted, but it also reports static inputs, an inventory that excluded surface-water flooding, and the need for retraining or fine-tuning in other regions. It is relevant to FloodSense's concerns with susceptibility and explanation, yet it differs fundamentally because FloodSense uses deterministic rules rather than machine-learning prediction. The model, features, and reported performance cannot be transferred to Bacoor City without local data, validation, and a different approved method.

Related Local Studies

GIS-Based Mapping of Flashflood Susceptible Areas in Bukang Liwayway, Kibawe, Bukidnon

Puno et al. (2022) generated a flashflood-susceptibility map for Barangay Bukang Liwayway, Kibawe, Bukidnon using GIS and frequency-ratio analysis. The researchers collected 160 flashflood inventory points and evaluated nine conditioning factors, including terrain, drainage, land cover, vegetation, and rainfall, using satellite, elevation, rainfall, and field information. They reported that 89.8% of the area fell within low, 4.4% within moderate, and 5.7% within high susceptibility, with 25 households exposed to the high category.

The study is similar to FloodSense in producing spatially organized susceptibility information and emphasizing local planning, but its statistical frequency-ratio method and rural geographic context differ from FloodSense's deterministic, user-requested scenario flow. Its sample points, factor relationships, weights, land-cover conditions, and proposed evacuation-site analysis are specific to Bukang Liwayway. They cannot be generalized to Bacoor City without locally appropriate data, expert review, field validation, and institutional authorization.

Flood Risk Assessment Using GIS-Based Analytical Hierarchy Process in the Municipality of Odiongan, Romblon, Philippines

Gacu, Monjardin, Senoro, and Tan (2022) assessed flood risk in Odiongan, Romblon using a GIS-based analytical hierarchy process under the Sendai Framework. The study combined five hazard, nine vulnerability, and three exposure parameters, obtained comparative judgments from ten experts and end-users, created weighted-overlay maps, and checked results against ground assessment and historical events. It identified nine barangays with high flood risk and reported that 3.26% of the municipality was in the high-risk class.

The work demonstrates a documented approach for combining spatial criteria, expert judgment, and validation. It is relevant to FloodSense's emphasis on geographic evidence and expert review, but it assesses risk by combining hazard, vulnerability,

and exposure rather than producing FloodSense's bounded susceptibility output. Its judgments and weights are subjective and locally derived, so they cannot be reused as Bacoor rules or parameters without renewed local evidence and authorization.

Web-Based Solution for Flood Warning Decision Support in the Province of Leyte, Philippines

Bentoso et al. (2021) developed a sensor-based engineering prototype intended to support flood-warning decisions in the Province of Leyte. The system collected rainfall and water-level readings, transmitted the data to a web interface, presented graphical information, and used thresholds to send warning messages; the authors also reported functional testing of the prototype. The documented output was a working monitoring and alert interface for responsible users, not evidence of province-wide operational deployment.

The Leyte study is relevant because it separates sensed conditions, decision support, and warning communication within a digital flood application. It differs from FloodSense because it relies on continuous sensors and alert thresholds, whereas FloodSense has no continuous monitoring, automatic polling, autonomous alerts, or official warning authority. Its sensor placement, communications, thresholds, and local results cannot be transferred to Bacoor City without new scientific, technical, and institutional validation.

Synthesis of the Reviewed Literature

The reviewed literature and studies show that flood susceptibility, vulnerability, hazard, exposure, and risk are related but not interchangeable. Membele et al. (2022) identified wide variation in mapping terminology, indicators, scale, participation, and validation, while Puno et al. (2022) and Gacu et al. (2022) demonstrated how local studies combine geographic factors, documented analytical procedures, and validation. Their methods support disciplined geographic assessment, but their factor

weights, classifications, and findings cannot be imported into Bacoor City because the study areas, data, scales, and intended outputs differ.

Mostafiz et al. (2022) showed that flood information is more useful when it supports understandable action, and official platforms such as UP NOAH and HazardHunterPH illustrate public geographic communication with visible institutional sources. FloodSense therefore separates its deterministic susceptibility result from preparedness guidance and identifies the scenario, location, source, and limitations presented to the user. Bentoso et al. (2021) demonstrates a different approach based on sensors and warning messages; its contribution is relevant to flood decision support but does not justify adding real-time monitoring or autonomous alerts to FloodSense.

Explainability and validation remain distinct responsibilities. Wang et al. (2025) showed that an interpretable machine-learning model can be evaluated quantitatively, but its static inputs, inventory limitations, and retraining requirement also show why performance in one area does not establish validity elsewhere. FloodSense instead uses deterministic rule matching so that the same approved facts and rule version produce the same result, yet repeatability alone does not validate the rules. Geographic provenance, local expert review, institutional authorization, and appropriate Bacoor data remain necessary.

Across the sources, geographic technologies support storage, analysis, visualization, and communication, but technical capability does not establish scientific or institutional authority. Administrative boundaries can support location resolution without containing flood-susceptibility facts. Official platforms, reviewed studies, and FloodSense also serve different users and disaster-management phases; therefore, similarities in mapping or decision support must not be interpreted as equivalence in data, functions, or authority.

Research Gap

Existing research already contributes flood-susceptibility mapping, GIS-based multicriteria assessment, explainable modeling, risk communication, sensor-based warning, and public hazard platforms. The inspected evidence nevertheless leaves a context-specific need for public-facing, location-specific, scenario-based susceptibility information that clearly explains a deterministic result and then presents preparedness guidance as a separate form of support. This is not a claim that FloodSense is the first flood application or that no comparable system exists.

The remaining gap also concerns provenance and transferability. The reviewed studies do not provide an approved set of Bacoor City susceptibility factors, rule thresholds, validated geographic inputs, or preparedness content that can be adopted automatically. External formulas, weights, model performance, sensor thresholds, and map classifications were developed for other data and places. FloodSense therefore requires local expert and institutional validation and remains a pre-event research prototype rather than an official forecasting, warning, evacuation, or hazard-certification system.

Comparative Synthesis of Related Studies

The following tables compare the verified methodologies, objectives, and contributions of the foreign and Philippine research studies retained in this chapter. Official platforms are discussed separately in the literature and existing-systems

discussion because their documentation does not automatically constitute a research methodology and results section.

Table 1 compares the retained foreign studies by author, verified research or technical procedure, objective, and documented contribution.

Table 1. Comparison of related foreign studies

SYSTEM	AUTHOR(S)/ YEAR	METHODOLOGY	OBJECTIVES	CONTRIBUTIONS
Examining Flood Vulnerability Mapping Approaches in Developing Countries: A Scoping Review	Membele, Naidu, and Mutanga (2022)	PRISMA-ScR scoping review of 130 studies selected from Web of Science and Scopus.	The review aimed to identify and compare flood-vulnerability mapping approaches used in developing countries.	It documented major variation in concepts, indicators, scales, participation, sensitivity analysis, and validation practice.
Actionable Information in Flood Risk Communications and the Potential for New Web-Based Tools for Long-Term Planning for Individuals and Community	Mostafiz, Rohli, Friedland, and Lee (2022)	Literature review of flood-risk communication and comparative review of 50 freely available web portals.	The study aimed to assess how flood information could better support long-term individual and community decisions.	It found limited micro-scale, actionable long-term planning information and identified needs for clearer decision support.

Table 1. Continued

SYSTEM	AUTHOR(S)/ YEAR	METHODOLOGY	OBJECTIVES	CONTRIBUTIONS
Accelerated and Interpretable Flood Susceptibility Mapping Through Explainable Deep Learning with Hydrological Prior Knowledge	Wang, Sanderson, Iqbal, and Woo (2025)	Comparative machine-learning and U-Net modeling with spatial five-fold cross-validation and Grad-CAM interpretation.	The study aimed to improve the speed, predictive performance, and interpretability of flood-susceptibility mapping.	The proposed model reported an AUC of 0.966 and faster convergence, while identifying data and transferability limits.

Table 2 compares the retained Philippine studies using the same evidence-based categories.

Table 2. Comparison of related local studies

SYSTEM	AUTHOR(S)/ YEAR	METHODOLOGY	OBJECTIVES	CONTRIBUTIONS
GIS-Based Mapping of Flashflood Susceptible Areas in Bukang Liwayway, Kibawe, Bukidnon	Puno et al. (2022)	GIS frequency-ratio analysis using 160 inventory points, nine conditioning factors, and field validation.	The study aimed to generate a localized flashflood-susceptibility map and identify exposed households and a possible evacuation site.	It mapped low, moderate, and high susceptibility and reported 25 households within the high-susceptibility area.
Flood Risk Assessment Using GIS-Based Analytical Hierarchy Process in the Municipality of Odiongan, Romblon, Philippines	Gacu, Monjardin, Senoro, and Tan (2022)	GIS-based analytical hierarchy process and weighted overlay using expert judgments, ground assessment, and historical-event checks.	The study aimed to map municipal flood risk by combining hazard, vulnerability, and exposure parameters.	It identified nine high-risk barangays and reported the mapped proportions of the municipality's risk classes.

Table 2. Continued

SYSTEM	AUTHOR(S)/ YEAR	METHODOLOGY	OBJECTIVES	CONTRIBUTIONS
Web-Based Solution for Flood Warning Decision Support in the Province of Leyte, Philippines	Bentoso et al. (2021)	Sensor-based engineering prototype using rainfall and water-level acquisition, web visualization, threshold alerts, and functional testing.	The study aimed to develop a web-based aid for monitoring flood conditions and supporting warning decisions in Leyte.	It reported graphical presentation of sensor readings and warning messages received by responsible authorities.

Table 3 compares documented functions of selected existing platforms and the Leyte research prototype. FloodAdapt (S1) and UP NOAH (S6) support flood-scenario or hazard-map exploration, while GloFAS (S2), IFIS (S3), and the Leyte prototype (S10) provide forecasts or sensor-based information. England's national flood-risk resource (S5) and HazardHunterPH (S7) support public geographic lookup, whereas GeoMapperPH (S8) and PlanSmartPH (S9) support governed institutional data or planning workflows. FloodSense combines a user-confirmed hypothetical rainfall scenario, a selected or foreground-confirmed location, deterministic rule-based susceptibility assessment, and separate preparedness guidance in a resident-initiated flow. The comparison does not imply that FloodSense provides live monitoring or forecasts or that its local susceptibility inputs are already validated.

Table 3. Comparison of developed study to the other existing systems.

Features	**Existing System**										**Proposed System**
	S1	**S2**	**S3**	**S4**	**S5**	**S6**	**S7**	**S8**	**S9**	**S10**	
Selectable flood scenario	✓					✓					✓
Interactive geographic display	✓	✓	✓	✓		✓	✓	✓			✓
Public location-based lookup			✓		✓	✓	✓				✓
Real-time monitoring or forecast		✓	✓							✓	
Rule-based susceptibility output											✓
Pre-event preparedness guidance					✓						✓
Governed administrative data workflow								✓	✓		✓
User-initiated foreground location											✓

Legend:

✓ - included in the documented system; blank - not established by the inspected source.

S1 - FloodAdapt: An Open-Source Application for Flood-Risk Scenario Modeling and Adaptation Planning

S2 - GloFAS: An Operational Platform for Global Flood Forecasting and Monitoring

S3 - IFIS: A Public Web Platform for Real-Time Flood Information, Sensor Data, and Forecasts

S4 - RAPT: A Geographic Information System for Community Resilience Planning and Data Visualization

S5 - National Assessment of Flood and Coastal Erosion Risk in England 2024: An Official Long-Term Risk Information Resource

S6 - UP NOAH Know Your Hazards and NOAH Studio: Interactive Philippine Platforms for Hazard Mapping and Flood-Scenario Visualization

S7 - HazardHunterPH: A Philippine Location-Based Application for Indicative Multi-Hazard Assessment

S8 - GeoMapperPH: A Governed Geospatial Application for Collecting Hazard, Exposure, Vulnerability, and Coping-Capacity Data

S9 - PlanSmartPH Ready to Rebuild: A Web-Based Rehabilitation and Recovery Planning Application for Local Government Units

S10 - Web-Based Solution for Flood Warning Decision Support in Leyte: A Sensor-Based Flood-Monitoring and Warning Prototype

METHODOLOGY

This chapter describes the research design, materials, requirements gathering, feasibility, participants, instruments, analysis plan, development evidence, testing, and planned evaluation for FloodSense. Completed activities, ongoing data collection, and planned work are distinguished throughout.

Materials

Research materials include the active approved resident questionnaire and privacy/consent notice, preliminary needs-assessment reports kept as separate historical records, BDRRMO and CPDCO interviews and letters, consultation records, official and scholarly sources, requirements documents, diagrams, interface evidence, and the 47-barangay administrative reference layer. Direct identifiers are excluded from public analysis.

Development and verification materials include Windows computers and available Android devices; Git version control; Flutter and Dart; Python, Django, Django REST Framework, and GeoDjango; PostgreSQL/PostGIS; Mapbox and OpenStreetMap-compatible mapping components; browser-based custom Admin templates; database migrations; API documentation; automated tests; and defect/verification records. Survey distribution and analysis use the exact approved instrument and an exported tabular dataset after the survey closes. Tools not supported by project records are not listed as methods.

Research Design

The study provisionally follows a developmental-descriptive research design. The developmental component covers the requirements-based creation and technical verification of the mobile, backend, geographic, Expert System, DSS, and administrative artifact. The descriptive quantitative component covers the resident questionnaire through frequencies and percentages, while qualitative key-informant interviews and document review explain institutional practice, data access,

provenance, and validation constraints. The final designation will be confirmed with the adviser before submission.

System Development Life Cycle

The software has been developed through successive requirements, design, implementation, testing, feedback, and correction increments. The available project records do not establish the roles and ceremonies required for Scrum, the maintained workflow and work-in-progress controls required for Kanban, or a documented RAD or Waterfall process. [AUTHOR NOTE: Select and describe a specific SDLC only after the group supplies the corresponding workflow evidence and obtains approval. Insert the final SDLC diagram here.]

Until that decision is made, Chapter 3 reports the actual activities rather than assigning an unsupported method name: requirements were elicited from surveys, interviews, documents, and consultations; functions were designed and implemented in backend, mobile, geographic, and Admin components; automated and manual checks informed corrections; and data-dependent or evaluation-dependent work remains pending.

Schedule Feasibility

The documented study period is September 2026 to May 2027. Major software functions are integrated, but authorized local-data acquisition, rule/content validation, survey closure and analysis, current backend verification, formal ISO/IEC 25010 evaluation, and manuscript completion remain on the critical path. A provisional Gantt chart is provided in Appendix [letter to confirm]. Its dates must be reconciled with the approved school calendar.

Cost Feasibility

The core development stack is open source, but map-provider usage, hosting, connectivity, transportation, printing, device testing, and deployment may create costs.

No reliable complete expense ledger was found. The line-item budget in Appendix [letter to confirm] therefore retains placeholders for receipts or approved estimates and excludes superseded Firebase or AI-assistant assumptions.

Operational Feasibility

The resident workflow is user initiated and separates assessment from guidance, while the custom Admin portal provides governed publication and record workflows. Operational adoption remains unconfirmed because the active needs survey and final evaluation are incomplete, approved susceptibility and center datasets are unavailable, and production ownership and support arrangements have not been established.

Technical Feasibility

The current repository establishes prototype-level technical feasibility for authentication, scenario/location input, deterministic assessment and explanation, geographic display, structured DSS guidance, nearest-center lookup, and governed Admin workflows. Current mobile verification passed 389 tests and static analysis. A current backend rerun, production-like deployment, security/performance/accessibility testing, and authorized operational data remain pending.

Requirements Gathering

Resident requirements are gathered through the active questionnaire on flood experience, information practices, timing, clarity, location specificity, consistency, decision effect, preparedness information, mobile access, and delivery preference. BDRRMO and CPDCO interviews inform institutional practices, source ownership, geographic/data access, evacuation information, and validation limitations. Literature and official documents inform concepts and safety boundaries. A proposed function becomes a thesis requirement only after it is traceable to evidence, feasible and safe, consistent with the approved scope, and validated through the appropriate technical

or stakeholder method. The traceability matrix is provided in Appendix [letter to confirm].

Participants and Sampling

The active survey targets consenting adults aged 18 or older who live in Bacoor City communities included in the study. The final sampling method, distribution denominator, response rate, and handling of Q2 ‘I am not sure’ require documentation. [Insert final number of valid respondents.] [Insert final distribution by barangay.] [Insert final excluded or invalid submissions and response rate.] Key informants include available BDRRMO and CPDCO representatives whose statements are used for requirements and limitations, not as proof that the application is effective.

Research Instruments

The active instrument is the bilingual FloodSense Resident Survey Questionnaire with a Data Privacy Act notice and consent, demographics, eligibility/flood experience, current flood-information experience, and technology access/preferences. It is preserved unchanged while responses are being collected. Interview guides, permission letters, consultation records, functional test cases, defect records, and the planned ISO/IEC 25010-based evaluation instrument support other evidence stages. Instrument validation records and the final evaluation form remain to be supplied.

Data Gathering and Analysis

After closure, the survey export will be archived with its cutoff and cleaned under documented consent, eligibility, duplicate, range, missing-data, and multiple-response rules. Categorical items will be summarized by counts and valid percentages; ordered items will be shown through distributions and medians where useful. Weighted means will not be used unless the actual scale and adviser-approved interpretation justify them. Barangay text will be normalized through a documented codebook. Interview and document evidence will be coded thematically

for information practices, data ownership, provenance, validation, governance, and scope. The item-level plan is provided in Appendix [letter to confirm].

Testing and Evaluation

Functional verification covers rule matching and conflict handling, validation, API responses, spatial queries, authentication and permissions, account flows, scenario/location confirmation, map behavior, structured DSS flow, nearest-center behavior, center import, and Admin workflows. Integration testing checks mobile-to-API and service-to-database behavior. Test records must identify case, precondition, input, expected result, actual result, status, environment, date, and defect link. Current mobile tests pass; the backend result available from project documents is historical until rerun.

Formal product-quality evaluation is planned using only the relevant ISO/IEC 25010:2023 characteristics and an approved instrument. The evaluator groups, sample, tasks, response scale, scoring, interpretation bands, consent, and analysis must be defined before administration. [Insert final evaluator count, scores, comments, and interpretation.] Automated test success and user ratings will be reported separately because neither substitutes for the other.

Ethical and Data-Governance Procedures

Participation is voluntary and governed by the active consent/privacy notice. Names and other direct identifiers will be removed from the analysis copy, small geographic cells will be reviewed for disclosure risk, and access will be limited to authorized researchers and academic advisers. Geographic, susceptibility, preparedness, and evacuation information will be labeled by source, validation, and publication status; technical processing alone will not be described as institutional approval.

Research Limitations

The active resident survey, final system evaluation, and several validation records are incomplete. The absence of an approved operational susceptibility dataset prevents official Bacoor classification claims. Nearest-center and DSS software functions do not establish that every underlying record is authorized or current. Assessment history, live alerts, fully offline use, fixed-schema Expert System parameter CSV exchange, official forecasting/warnings, and production deployment are not established.

[AUTHOR NOTE: Complete the survey-dependent findings and final evaluation sections only after the validated exports, cleaning log, approved analysis, and interpretation are available.]

The provisional appendices contain the active questionnaire reference, traceability materials, feasibility tables, diagrams, and testing/evaluation templates. Appendix letters will be finalized after the complete order is approved.

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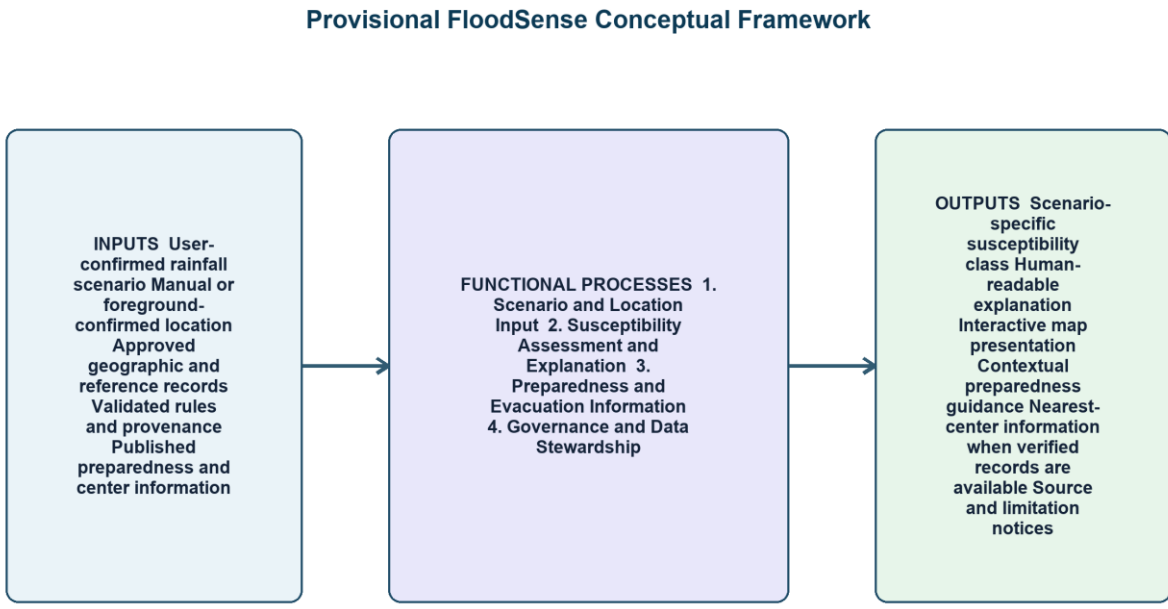
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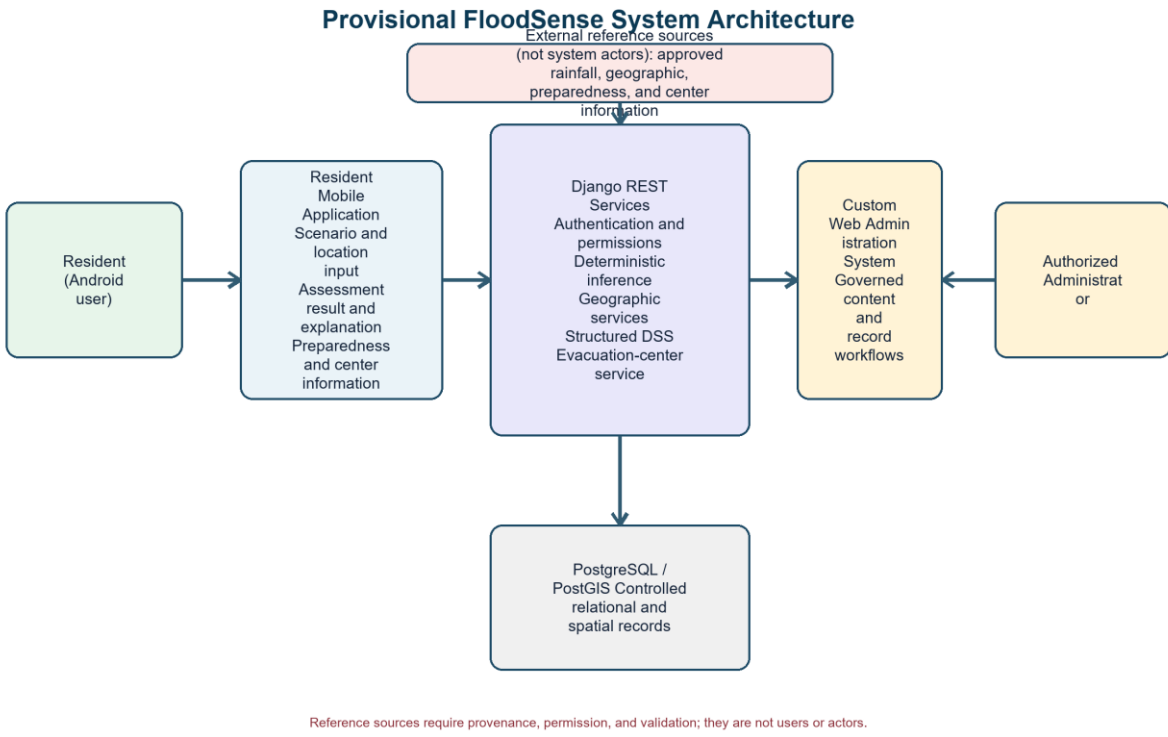
PROVISIONAL APPENDICES (ORDER AND LETTERS TO BE CONFIRMED)

Appendix [Letter to Confirm]. Provisional Conceptual Framework



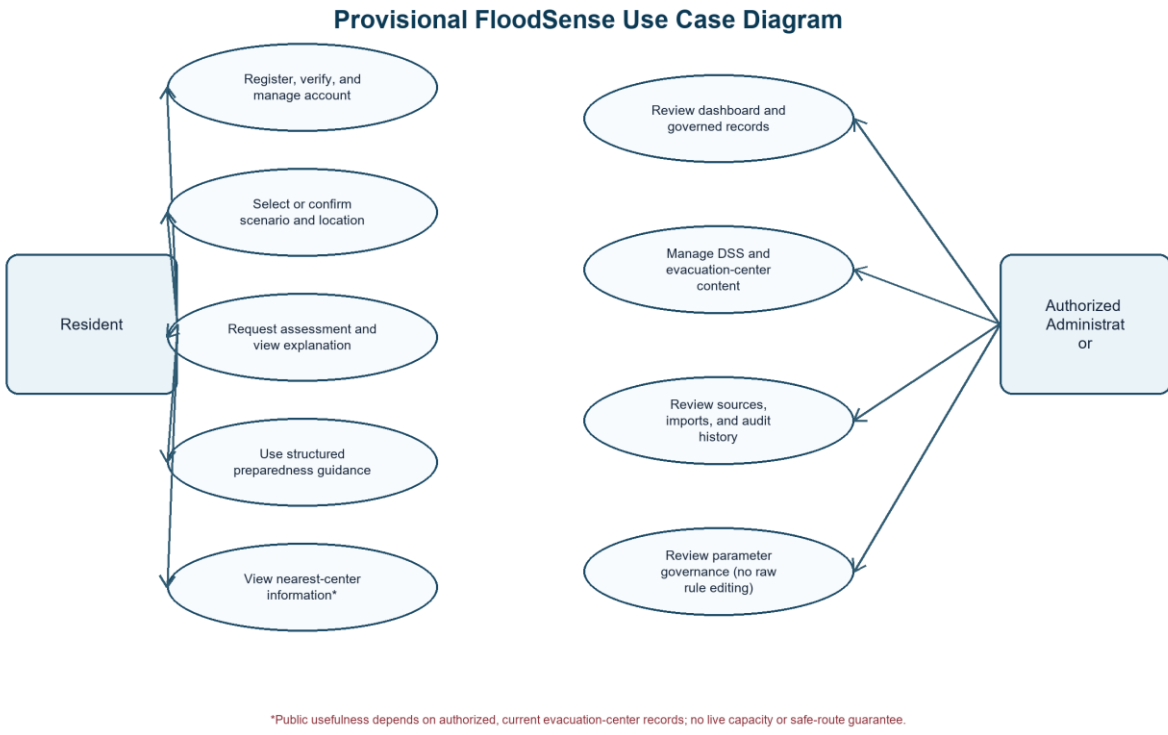
Provisional: validate actors, data sources, relationships, notation, and appendix numbering before final submission.

Appendix [Letter to Confirm]. Provisional System Architecture



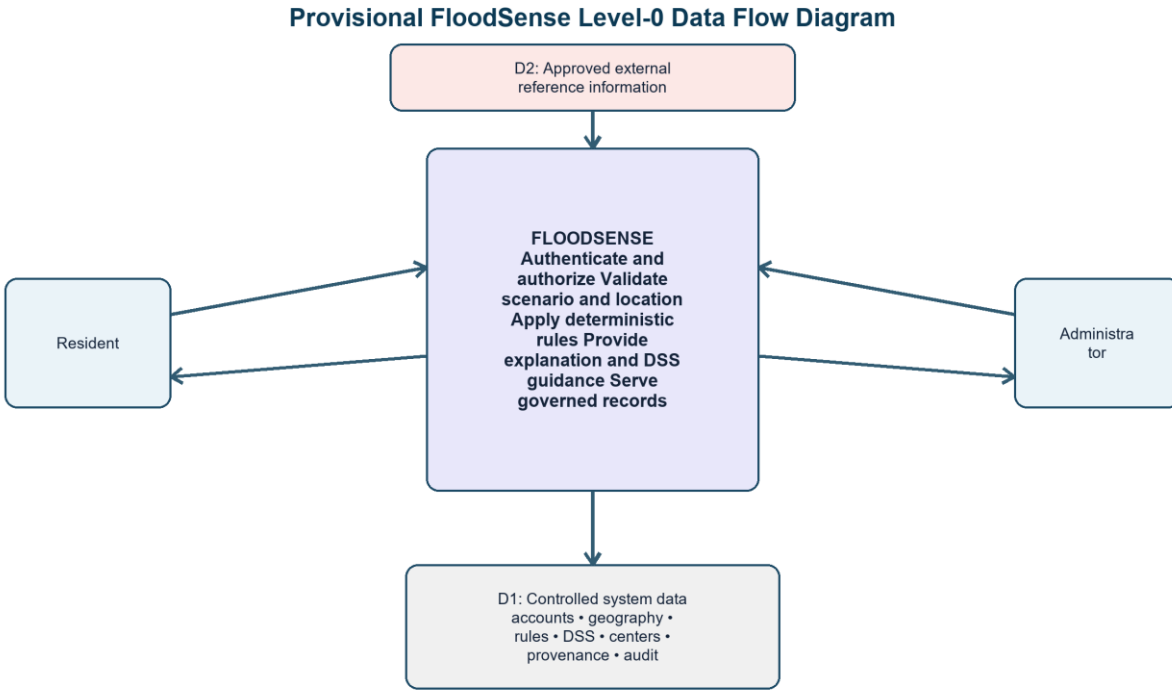
Provisional: validate actors, data sources, relationships, notation, and appendix numbering before final submission.

Appendix [Letter to Confirm]. Provisional Use Case Diagram



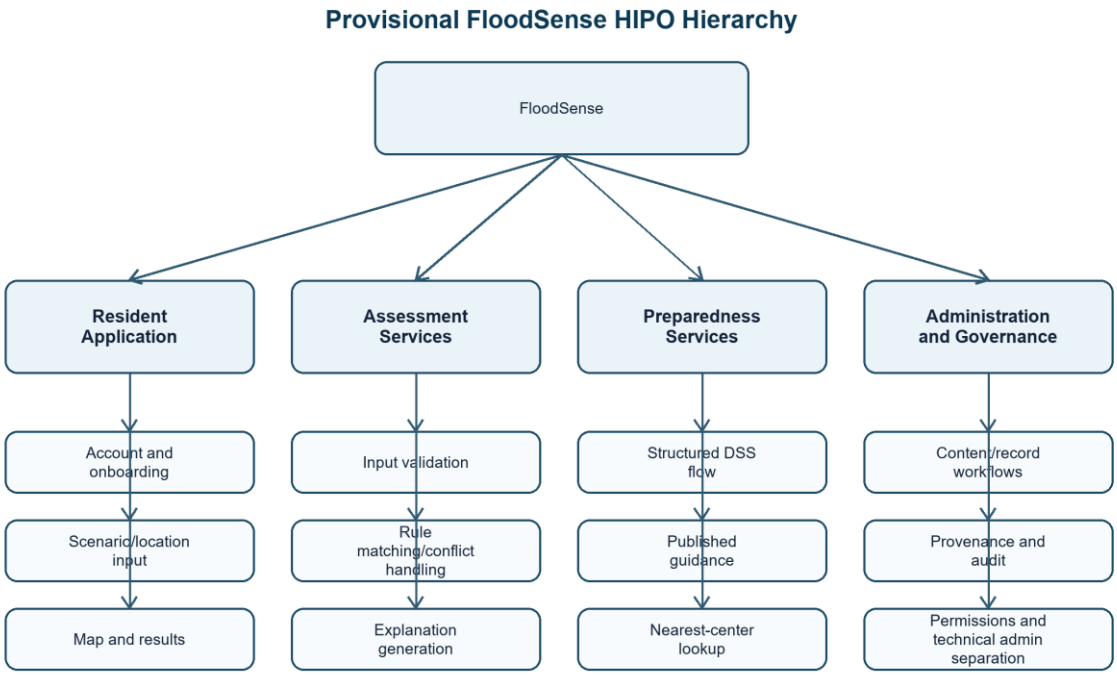
Provisional: validate actors, data sources, relationships, notation, and appendix numbering before final submission.

Appendix [Letter to Confirm]. Provisional Level-0 Data Flow Diagram



Provisional: validate actors, data sources, relationships, notation, and appendix numbering before final submission.

Appendix [Letter to Confirm]. Provisional HIPO Diagram



Provisional: validate actors, data sources, relationships, notation, and appendix numbering before final submission.

Appendix [Letter to Confirm]. Schedule Feasibility and Gantt Chart

See the phase schedule in 05_Feasibility_Study_Materials.md. Replace provisional periods with the approved academic calendar and actual completion records.

Appendix [Letter to Confirm]. Cost Feasibility and Line-Item Budget

See the cost template in 05_Feasibility_Study_Materials.md. Enter only documented receipts or approved estimates before calculating the total.